

Week 2

Assignment : Computation with Variables & Formulas

192-101 Computer Programming with Generative AI · 3 problems ·

Hand-written only. Write each answer **BY HAND** on this sheet.

No AI. Using AI assistants or code generators for this assignment is prohibited.

HOW TO ANSWER : READ THE BANNER ON EACH PROBLEM

- **Algorithm** — numbered steps in plain English (the plan).
- **Pseudocode** — code-like steps using READ / SET / PRINT (language-independent).
- **Flowchart** — boxes and arrows: oval = start/end, parallelogram = input/output, rectangle = process.
- **Code** — Python. Allowed: variables, input(), int()/float(), + - * // % **, f-strings, and
- import math (**sqrt**, sin, cos, asin, radians, pi).

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Problem 1 : How Far Apart Are Two Points?

YOUR TASK: WRITE THE CODE

Read the coordinates of two points, (x1, y1) and (x2, y2). Compute and print three things: the straight-line (Euclidean) distance, the grid (Manhattan) distance, and the midpoint between them.

Real use: game hit-detection and physics use straight-line distance; grid / tile movement and routing use Manhattan distance; the midpoint centres a camera or places a label.

YOUR PROGRAM MUST

- read four numbers: x1, y1, x2, y2
- Euclidean distance = $\sqrt{(x2-x1)^2 + (y2-y1)^2}$
- Manhattan distance = $\text{abs}(x2-x1) + \text{abs}(y2-y1)$
- midpoint = $((x1+x2)/2, (y1+y2)/2)$

SAMPLE RUN

```
Enter x1: 2  Enter y1: 3
Enter x2: 10 Enter y2: 9
Euclidean distance: 10.00
Manhattan distance: 14.00
Midpoint: (6.00, 6.00)
```

Hint: `abs(...)` gives a positive value; a square root is `value ** 0.5` or `math.sqrt(value)`.



```
import math
```

```
# Read the four coordinates
```

```
x1 = float(input("Enter x1: "))
```

```
y1 = float(input("Enter y1: "))
```

```
x2 = float(input("Enter x2: "))
```

```
y2 = float(input("Enter y2: "))
```

```
# Straight-line distance
```

```
euclidean = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
```

```
# Grid distance
```

```
manhattan = abs(x2 - x1) + abs(y2 - y1)
```

```
# Midpoint
```

```
mid_x = (x1 + x2) / 2
```

```
mid_y = (y1 + y2) / 2
```

```
print(f"Euclidean distance: {euclidean:.2f}")
```

```
print(f"Manhattan distance: {manhattan:.2f}")
```

```
print(f"Midpoint: ({mid_x:.2f}, {mid_y:.2f})")
```

```
... Enter x1: 4.5
Enter y1: 3.5
Enter x2: 7.9
Enter y2: 3.9
Euclidean distance: 3.42
Manhattan distance: 3.80
Midpoint: (6.20, 3.70)
```

Problem 2 : Area of a Triangle From Its Corners

YOUR TASK: WRITE THE CODE

A triangle is given by its three corner points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) . Print its perimeter (total side length) and its area.

Real use: this is the “shoelace” area formula used in graphics, CAD, and GIS to measure any polygon — e.g. a plot of land or a shape on screen — directly from its corner coordinates.

YOUR PROGRAM MUST

- read six numbers for the three corners
- find each side length with the distance formula, then add them for the perimeter
- $\text{area} = \text{abs}(x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)) / 2$

SAMPLE RUN

```
Corner 1 (x y): 0 0
Corner 2 (x y): 4 0
Corner 3 (x y): 4 3
Perimeter: 12.00
Area: 6.00
```

Hint: the three side lengths are the distances $P_1 \rightarrow P_2$, $P_2 \rightarrow P_3$, and $P_3 \rightarrow P_1$. Reuse the distance idea from Problem 1.

```

import math

# Read the three corner points
print("Corner 1 (x y): ", end="")
x1, y1 = map(float, input().split())

print("Corner 2 (x y): ", end="")
x2, y2 = map(float, input().split())

print("Corner 3 (x y): ", end="")
x3, y3 = map(float, input().split())

# Calculate the three side lengths\
side1 = math.sqrt((x2 - x1)**2 + (y2 - y1)**2) # P1 → P2
side2 = math.sqrt((x3 - x2)**2 + (y3 - y2)**2) # P2 → P3
side3 = math.sqrt((x1 - x3)**2 + (y1 - y3)**2) # P3 → P1

# Perimeter = sum of the three sides
perimeter = side1 + side2 + side3

# Area using the shoelace formula
area = abs(x1*(y2 - y3) + x2*(y3 - y1) + x3*(y1 - y2)) / 2

# Print results
print(f"Perimeter: {perimeter:.2f}")
print(f"Area: {area:.2f}")

```

```

Corner 1 (x y): 1 2
Corner 2 (x y): 4 5
Corner 3 (x y): 2 6
Perimeter: 10.60
Area: 4.50

```

Problem 3 : Brightness & Hex Code (Colour)

YOUR TASK: READ THE CODE, THEN WRITE THE ALGORITHM AND THE PSEUDOCODE

The working Python program below reads a colour as three whole numbers r, g, b (0–255) and prints its perceived brightness (Rec. 709 luminance) and its web hex code. Study it, then (a) write the algorithm as numbered plain-English steps, and (b) write the equivalent pseudocode. Do not rewrite Python.

Real use: developers convert RGB to hex constantly, and the brightness value decides whether black or white text sits on a coloured button — the core of accessibility contrast checks.

GIVEN PROGRAM (PYTHON)

```
r = int(input("Enter R: "))
g = int(input("Enter G: "))
b = int(input("Enter B: "))
brightness = 0.2126 * r + 0.7152 * g + 0.0722 * b
hex_code = f'#{r:02X}{g:02X}{b:02X}'
print(f"Brightness: {brightness:.2f}")
print(f"Hex code: {hex_code}")
```

WHAT IT PRINTS FOR R=255, G=140, B=0

```
Brightness: 154.34
Hex code: #FF8C00
```

Hint: the algorithm is “what it does” in order (read the three values, combine them, format, print); the pseudocode says the same using READ / SET / PRINT.

(a) Algorithm - numbered steps in plain English (b) Pseudocode

(a) Algorithm

1. Ask the user to enter the red value (R), the green value (G), the blue value (B) and store it as a whole number.
2. Calculate the brightness using the formula:

$$\text{brightness} = 0.2126 \times R + 0.7152 \times G + 0.0722 \times B$$

3. Create the hex color code by converting R, G and B into two-digit hexadecimal numbers and joining them with a “#” in front.
4. Print the brightness (rounded to 2 decimal places).
5. Print the hex color code.

(b) Pseudocode

BEGIN

 READ R

 READ G

 READ B

 SET brightness $\leftarrow 0.2126 * R + 0.7152 * G + 0.0722 * B$

 SET hex_code $\leftarrow \text{"#"} + \text{HEX}(R) + \text{HEX}(G) + \text{HEX}(B)$

 PRINT "Brightness: " + brightness (2 decimal places)

 PRINT "Hex code: " + hex_code